25900013 **Reg. No:……**
Name:……..

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBIF100- BIOLOGICAL DATA AND MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A
Short Answer Questions
Answer any **5** questions
Each question carries **2** Marks

1. Describe two advantages of using a Database Management System (DBMS). [U]/[CO1]

- Data redundancy is minimized through centralized control.
- Ensures data integrity and security with access control mechanisms.

2. Articulate two features of OMIM. [A]/[CO4]

- Contains detailed gene-phenotype relationships.
- Updated regularly with curated genetic disorder information.

3. Demonstrate nature of biological data. [U]/[CO2]

- Highly heterogeneous (e.g., DNA, RNA, protein sequences).
- Large-scale and complex, requiring structured storage systems.

4. Discuss classification of biological database. [U]/[CO2]

- Primary databases: Contain raw experimental data (e.g., GenBank).
- Secondary databases: Derived data with annotations (e.g., Prosite, InterPro).

5. Prepare a list of Literature databases. [A]/[CO4]

- PubMed
- Scopus
- Web of Science

- Google Scholar

6. Discover the role of DDBJ in sequence storage. [A]/[CO4]

- Stores nucleotide sequences from global researchers.
- Part of the INSDC (with GenBank and EMBL).

7. Report an interesting feature of Prosite. [A]/[CO4]

- Identifies protein families using conserved motifs and patterns.

[2*5=10]

PART B

Short Essay Questions
Answer any **5** questions
Each question carries **5** Marks

8. Compare relational databases and NoSQL databases. [U]/[CO1]

- Relational: Structured tables, use SQL, ideal for structured data.
- NoSQL: Schema-less, supports documents/key-value pairs, good for unstructured data.
- Scalability: NoSQL is better for horizontal scaling.
- Flexibility: NoSQL adapts to changes faster.
- Use case: RDBMS for banking; NoSQL for social media.

9. Compare and contrast physical and logical database schemas. [U]/[CO1]

- Physical Schema: Deals with how data is stored physically (e.g., indexing, partitions).
- Logical Schema: Abstract design (e.g., table structure, relationships).
- Physical changes don't affect logical design.
- Logical schema is more user-focused; physical is system-focused.
- Together, they define a complete data model.

10. Describe the historical development of biological databases. [U]/[CO2]

- 1970s: Start of GenBank and EMBL.
- 1980s: SWISS-PROT and OMIM launched.
- 1990s onwards: Growth of specialized databases like KEGG, DDBJ.
- Emergence of NCBI as an integrative platform.
- Shift from flat files to relational systems and web interfaces.

11. Discuss how data collected and stored in primary databases [U]/[CO3]

- Submission by researchers post-publication.
- Includes metadata, raw sequences, experiment details.
- Standardized formats (FASTA, GenBank).
- Peer-reviewed and curated.
- Accessible to public via web portals.

12. Illustrate a SWISS-PROT entry format. [A]/[CO4]

- ID: Accession number
- DE: Description of protein

- GN: Gene name
- OS: Organism source
- DR: Cross-references
- SQ: Sequence data with checksum

13. Examine the features of a genome database entry. [A]/[CO4]

- Complete nucleotide sequence.
- Annotations: gene locations, exons, regulatory regions.
- Cross-references to protein and expression data.
- Organism-specific metadata.
- Visualization tools for genomic browsing.

[5*4=20]

PART C

Long Essay Questions
Answer any **2** questions
Each question carries **10** Marks

14. Relate the importance of data consistency, isolation, and durability in a database transaction. Illustrate with examples. [U]/[CO1]

- Atomicity: Entire transaction succeeds or fails.
- Consistency: Database moves from one valid state to another.
- Isolation: Transactions occur independently.
- Durability: Committed data remains safe post-failure.
- Ensures data reliability and integrity in bioinformatics databases.(6 marks)
- Examples: Banking systems, sequence updates in NCBI.(4 marks)

15. Discuss how data trends have evolved over the years and their impact on decision-making in various industries. [U]/[CO1]

- Shift from structured to unstructured big data.
- Rise of real-time analytics, cloud computing.
- AI and machine learning for pattern discovery.
- Enhanced predictive modeling in healthcare, agriculture, and genomics.
- Data-driven decision-making now standard in R&D.

16. Discuss the significance of NCBI in bioinformatics and its contributions to biological research. [U]/[CO2]

- Significance- Hosts GenBank, PubMed, BLAST tools.
- Integrates genomic, literature, and protein data.
- Supports open access to biological data.(5 marks)
- Importance- Encourages standardization and curation.
- Major hub for bioinformatics research and education.(5 marks)

17. Articulate the importance of compound databases. [A]/[CO4]

- Definition- Store chemical and biochemical compound data.
- Crucial for drug design and toxicology studies. (3 marks)

- Examples: PubChem, ChEBI. (2 marks)
- Importance- Enable structure searches and activity predictions.
- Integrated with genomics and proteomics for holistic analysis.(5 marks)

[10*2=20]